# Tumour classification — results and progress

Sheet for a project guide. Research scores on UM-BMID phantoms, not a clinical diagnosis. Formulas, cut-offs and the step-by-step test are in how_classification_works.pdf.

## 1. Where the work stands

A first scan, with no earlier healthy scan of the same breast, scores **71 / 100** (71 ± 2 across 3 nested repeats). A follow-up that subtracts earlier healthy scans of the same breast scores **97 / 100**. The first-scan number has not been lowered. A later side check scored about 69 because it was a smaller test: signal features only, and training scans from the fair set only. The official model still uses signal plus image, and it may train on every other scan of phantoms that are not in the test fold.

| Question | Official result | What was compared |
|---|---|---|
| Tumour on a first scan | **71 ± 2** | Recording details only 47 / 100; chance 50 |
| First scan, older / newest phantoms | 67 / 72 | Same nested test, split by generation |
| First-scan alarm (cut-off from training phantoms) | 66% found / 65% cleared | Default 0.5 cut is less balanced |
| Tumour on a follow-up | **97 ± 0** | Older 86, newest 99, ≤2 cm 97 |
| Follow-up alarm | 88% found / 95% cleared | Same number of earlier healthy scans for both classes, at most 3 |
| Size class, first scan / follow-up | 64% / 56% | Chance 33%. Cuts at 2 cm and 4 cm |
| Diameter error, first scan / follow-up | 0.7 cm / 0.4 cm | Newest follow-up phantoms 0.2 cm |
| Side check: signal only, then signal + formulas | 69 ± 2 then 68 ± 2 | Does not replace the official first-scan row |
| Criticality, first scan (volume index) | 71 ± 3 | Chance 33. Not clinical BI-RADS |

## 2. Progress since the reconstruction-only stage

• Detection and characterisation now have a fair test: equal tumour and healthy counts inside a session, and no phantom in both training and test.

• The saved first-scan model is signal + image, random forest, trained on fair scans plus all other scans. Its yes/no cut-off is the Youden point (highest true-positive rate minus false-positive rate) on training phantoms only.

• The saved follow-up model is signal (empty chamber + same-from-every-angle part removed), logistic, trained on fair scans only. The reference is the mean of other healthy scans from the same sitting, and the tumour is never given more of those scans than the healthy cases.

• Things tried for the first scan that did not raise the official score: pasting a measured tumour echo onto other training breasts, subtracting an average fat shell from other breasts, and adding the scattering-spectrum and frequency-slope formulas. The follow-up score stayed near 98 when the formulas were added.

• A convolutional network on the breast-window sinogram scored 50/100 on this fair test, and a PCA-then-logistic classifier on that sinogram scored 45/100. The network fit the training breasts and did not carry that fit to a new phantom. Neither was adopted. Details are in how_classification_works.pdf, Step 59.

• Type, meaning sphere versus spiculated, is not reported as a classifier. Only 6 of the fair tumour scans are spiculated. Criticality is reported instead, from diameter and that shape flag.

• Closed-form radar location (round-trip delay of the brightest antenna) misses by more than the coherent image spot, so position stays with the image.

## 3. Comparison with published studies

Protocols differ, so this is context, not a ranking. Reimer and Pistorius, IEEE J-ERM 2022, report a CNN at 78 ± 3 on a leave-one-fat-shell-out split of this dataset, and 90 ± 3 only when the test breast is inside the training volume and the tumour is at antenna height. Their logistic regression did not generalise on that split. The EuCAP 2020 accuracy of 85 ± 4 was a subset. Davis et al. 2008 and Conceição et al. 2010 report size and shape accuracies above 70–97 percent on simulated tumour echoes, without a measured skin and fat shell. The 2023 review gives image-based sensitivity 63–100 percent and specificity 20–65 percent in the few studies that stated it.

What those papers do not report is the follow-up number under a matched reference count: 97 / 100. The first-scan number here is the stricter nested score, not a claim to have beaten the 2022 network on its own split.

## 4. Limits to state with the numbers

• 27 phantoms in the first-scan test and 24 in the follow-up test. The ± is the spread across repeats.

• No benign versus malignant label. Metadata BI-RADS is breast density. Criticality is a research volume index.

• Generation 2 is excluded: tumour and healthy scans were taken in different sittings, and the sitting number alone gives the answer.

• Large 6 cm tumours exist only in the older phantoms. The newest phantoms have four tumour positions.

Numbers: results/classification/final_summary.json. Process note: results/reports/classification/how_classification_works.pdf.